



Align-then-Aggregate: Pretreatment prediction of radiation-induced lymphopenia in nasopharyngeal carcinoma via multi-modal collaborative learning

Rongchang Zhao^{a,*}, Hengzhang Deng^a, Zhangyue Wu^a, Zijian Zhang^b

^a School of Computer Science and Engineering, Central South University, Changsha, China

^b National Clinical Research Center for Geriatric Disorders, Xiangya Hospital, Central South University, Changsha, China

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ABSTRACT

Pretreatment prediction of radiation-induced lymphopenia (RIL) aims to proactively assess the risk of lymphopenia and improve post-radiotherapy survival, particularly for patients with nasopharyngeal carcinoma. Existing methods are typically based on expert-designed workflows, where handcrafted features are independently extracted from computed tomography (CT) images, dose maps, and clinical factors, and directly combined to predict RIL grades. These methods are subject to human bias, neglect inter-modal heterogeneity and lack effective modeling of complementary inter-modal interaction, which limits the performance of RIL prediction. Based on this, a novel multi-modal collaborative learning framework **Align-then-Aggregate Network (AANet)** is proposed to achieve accurate RIL prediction in an end-to-end manner using an Align-then-Aggregate strategy, where the alignment stage maps CT and dose features into hierarchically aligned representations that are spatially co-registered and semantically comparable, facilitating the extraction of discriminative patterns to RIL, and the aggregation stage enables modality-specific information exchange guided by patient clinical information, allowing inter-modal complementarity and providing a more comprehensive representation of the patient. Compared to previous methods, AANet can fully automatically extract features from multiple modalities and effectively integrate heterogeneous modalities using a two-stage Align-then-Aggregate strategy and multi-perspective fusion algorithm, thereby improving RIL prediction. Extensive experiments on a newly established dataset demonstrate that AANet achieves accurate RIL prediction with an accuracy of 0.8148 ± 0.0380 and an area under the curve of 0.8562 ± 0.0450 , outperforming existing methods.

1. Introduction

Predicting radiation-induced lymphopenia (RIL) on the basis of pre-RT factors is significant for improving radiation therapy (RT) planning [1,2]. RIL is a type of inevitable toxic effect in RT, which refers to a significant decrease in absolute lymphocyte count (ALC) in the blood after radiation exposure. Severe RIL leads to inferior overall survival, progression-free survival, and distant metastasis-free survival in patients. Many studies [3–5] have shown that a higher incidence of RIL exists in patients with head and neck cancer, e.g., nasopharyngeal carcinoma (NPC). Therefore, identifying patients with NPC who are at high risk of severe lymphopenia is essential, which may help them benefit from RIL mitigation strategies and ultimately improve their survival.

However, accurate RIL prediction is challenging because of the complex and dynamic biological mechanisms behind the decrease in ALC. Existing evidence suggests that RIL is jointly influenced by multiple

factors: (1) dosimetric exposure to lymphoid organs and organs with abundant blood circulation, which determines the extent of radiation-induced lymphocyte damage [6]; (2) patient-specific clinical factors, such as lower baseline ALC, older age, and higher tumor stage with poor physical fitness, which are associated with increased RIL risk [7, 8]; and (3) genetic susceptibility, which further contributes to inter-patient variability [9]. Together, these complex factors make it difficult to model the mechanisms underlying RIL in patients.

A number of classical methods have been proposed for RIL prediction, including mechanistic and learning-based approaches: (1) mechanistic methods model dose–lymphocyte depletion with simplified mathematical formulations. For example, Jin et al. [10] proposed a system of ordinary differential equations (ODEs) to characterize the dose–lymphocyte depletion relationship in each compartment. (2) learning-based methods leverage case data to learn mappings from CT, dose, and clinical indicators to lymphocyte reduction. For instance, Xu et al. [11]

* Corresponding author.

E-mail addresses: zhaorc@csu.edu.cn (R. Zhao), more_king@csu.edu.cn (H. Deng), 224712180@csu.edu.cn (Z. Wu), wanzjz@csu.edu.cn (Z. Zhang).

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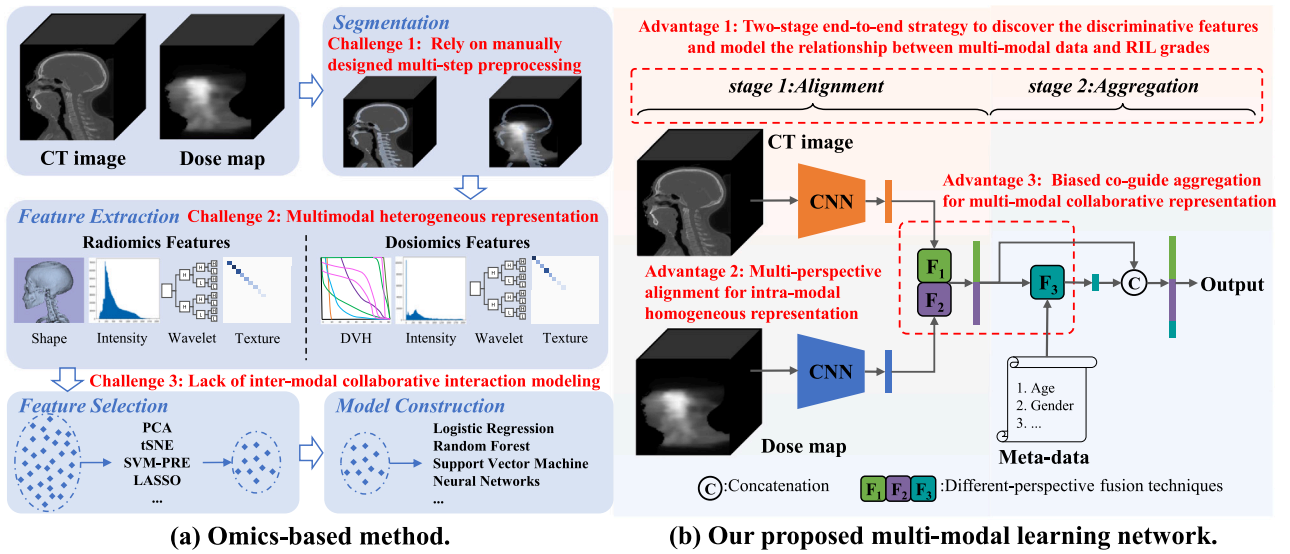


Fig. 1. (a) The challenges for RIL prediction. (b) Overview of the proposed AANet. It consists of two stages based on the Align-then-Aggregate strategy: alignment for eliminating inter-modal heterogeneity and aggregation for realizing inter-modal interactions.

proposed a multivariable logistic regression model using ECOG, pre-RT lymphocytes, pre-RT hemoglobin, dose-volume histogram (DVH), and other indicators to predict lymphocyte injury severity, and Zheng et al. [12] integrated dose distributions, DVH parameters, and clinical factors into a deep learning dosimetrics model for multi-center grade-4 RIL prediction. Compared with mechanistic approaches, learning-based methods can largely reduce reliance on expert experience, but still fail to model the complex influence of diverse factors on RIL grades.

Multi-modal learning can potentially improve RIL prediction by jointly leveraging CT, dosimetric, and clinical information. To address RIL prediction, existing methods still suffer from several drawbacks (Fig. 1(a)): (1) **Manually designed multi-step preprocessing introduces human bias.** Many methods rely on expert-designed multi-step preprocessing (e.g., ROI-based segmentation) and handcrafted descriptors such as DVH and GLCM, which may introduce human bias and limit performance [11,13]. (2) **Significant modality heterogeneity leads to catastrophic feature representation.** CT, dose maps, and clinical variables are highly heterogeneous in semantics and representation space, making naive fusion prone to inequitable representations and degraded prediction [14–16]. (3) **Lack of inter-modal collaborative interaction modeling.** Existing approaches often fuse modalities in a shallow manner (e.g., feature concatenation followed by a fully connected predictor), restricting cross-modal interactions and limiting the effective mapping from multi-modal features to RIL grades. [11,17].

In this paper, we propose a novel automated framework **Align-then-Aggregate Net (AANet)** to predict RIL grade end-to-end with multi-modal collaborative learning (Fig. 1(b)). Compared with existing methods, the proposed AANet effectively resolves intermodal heterogeneity and fully models the interactive relationships between modalities, which consists of two stages: **Aligning** CT images and dose maps explicitly at both spatial and semantic levels, while implicitly learning depth level alignment of features between imaging and clinical modalities for reducing modality heterogeneity, and **Aggregating** multi-modal features under the guidance of metadata to model the relationship between multi-modal data and RIL grades. Extensive experiments and results demonstrate that it achieves accurate RIL prediction with the proposed Align-then-Aggregate strategy, with an accuracy of 0.8148 ± 0.0380 and area under the curve of 0.8562 ± 0.0450 .

Overall, the main contributions are summarized as follows:

- (1) A novel multi-modal collaborative learning framework is proposed for the first time to achieve RIL prediction before radiation

therapy by modeling the inherent interactions between multi-modal data (CT images, dose maps, and clinical data) and a decrease in absolute lymphocyte count (Section 3).

- (2) A novel and effective Align-then-Aggregate strategy is designed to collaboratively learn lymphopenia-related features for RIL prediction by fusing the aligned multi-modal data with the guidance of metadata (Section 3.1).
- (3) Novel modules are proposed for hierarchical multi-modal interaction learning, by integrating semantic-level CT-dose alignment (Section 3.2), multi-scale spatial alignment (Section 3.3) and meta-guided biased cross-modal aggregation (Section 3.4).
- (4) Extensive experiments are conducted on a newly established dataset to evaluate the effectiveness of RIL prediction, and results show the advantages of our proposed method (Section 4).

In this work, we advance our preliminary attempt on RIL prediction [18] in the following aspects: (1) refine the Divide and Conquer approach into the Align-then-Aggregate framework to further improve the handling of heterogeneous multi-modal data; (2) break down the modal fusion process into multiple levels to achieve full integration; (3) carry out more extensive experiments on performance analysis and comparisons.

2. Related work

2.1. Radiation-induced lymphopenia prediction

RIL prediction is clinically important, and a variety of methods have been proposed in the literature. Existing studies can be broadly categorized into two lines. (1) **Mechanistic modeling.** Using simple mathematical models or sophisticated machine learning models, these methods typically model the association between carefully selected clinical or dosimetric predictors and RIL outcomes, aiming to identify key determinants with relatively limited data and high interpretability. For example, van Rossum et al. [19] analyzed factors such as age, planning target volume, body mass index (BMI), radiation modality, and baseline ALC for grade-4 RIL risk and developed a pretreatment clinical nomogram for esophageal cancer patients. Zhu et al. [20] identified individualized risk profiles for RIL using machine learning, while Yu et al. [21] investigated potential determinants via explainable machine learning in breast cancer cohorts. Huang et al. [13] further verified the importance of specific anatomical regions (e.g., cervical

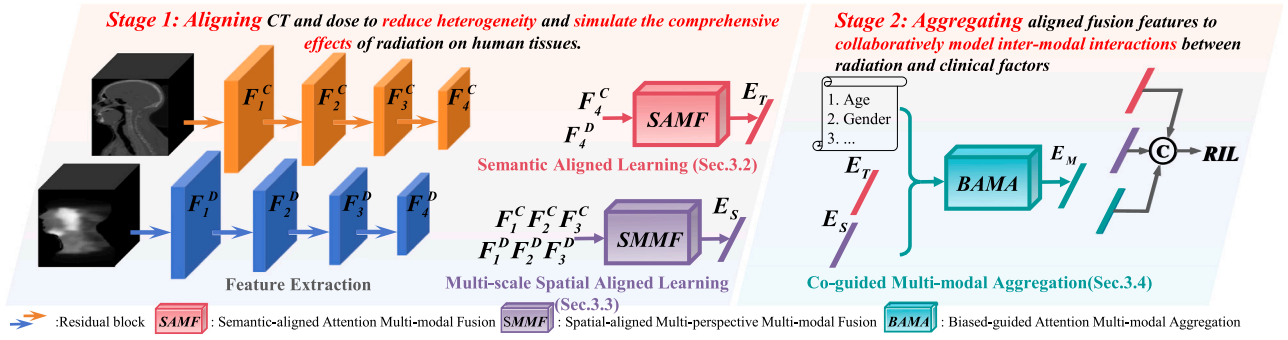


Fig. 2. Architecture of the proposed AANet. It employs three modules for multi-modal collaborative RIL prediction: the SAMF and SMMF modules in the first stage perform semantic and spatial alignment, respectively, and the BAMA module in the second stage aggregates the aligned modalities for joint RIL prediction.

spine and skull) for RIL in nasopharyngeal carcinoma using dosiomics and radiomics models. Despite their interpretability, these approaches are usually constrained by expert-designed pipelines and handcrafted predictors, making model construction labor-intensive and sensitive to variations that violate the assumed feature-to-outcome mapping, which may lead to degraded prediction and expensive re-modeling efforts.

(2) Learning-based prediction. In contrast, learning-based approaches design trainable deep networks to improve feature extraction and RIL prediction in a data-driven manner, leveraging patterns learned from larger cohorts. For example, Ebrahimi et al. [22] proposed a hybrid deep learning model in a stacked structure to predict a trend toward ALC depletion based on the clinical information before or at the early stages of RT treatment. Kim et al. [23] developed an ensemble model based on neural networks to predict RIL following concurrent chemotherapy for non-small cell lung cancer from both institutions. Zhu et al. [17] formulated RIL grade prediction as a classification problem and proposed a hybrid deep learning model. More recently, Zheng et al. [12] proposed deep learning dosiomics models built upon radiation dose distributions and further combined them with DVH parameters and clinical factors for grade-4 RIL prediction in a multi-center setting. Compared with knowledge-driven models, learning-based methods reduce reliance on expert experience and can learn more complex mappings from higher-dimensional patient data. However, existing learning-based predictors still tend to rely on incomplete factor coverage and often lack sufficient modeling of complementary collaboration among CT, dosimetric, and clinical cues, leaving room for more effective cross-modal interaction learning.

2.2. Multi-modal fusion for medical prediction

Multi-modal fusion has the potential to improve medical prediction by leveraging complementary information distributed across different modalities. In radiotherapy-related applications, Lu et al. [14] suggested a multi-modal information fusion strategy for prognosis forecasting in re-irradiation. Nevertheless, existing RIL-related multi-modal attempts are often still constrained by expert-designed pipelines (e.g., handcrafted features) and limited cross-modal interaction modeling, which may introduce human bias and restrict efficient utilization of complementary factors.

Beyond RIL prediction, multi-modal medical prediction has seen rapid technical progress in recent years, where models are designed to learn end-to-end task-relevant representations and explicitly model cross-modal interactions. For example, Zhang et al. [24] proposed an attention-based multi-modal fusion transformer to integrate pathology slides, MRI, and clinical information for treatment response prediction. Song et al. [25] further employed transformer-based cross-modality and cross-region attention to fuse CT and whole-slide pathology for outcome prediction. Shi et al. [26] integrated multiple clinical data modalities, including histopathology, radiology, endoscopy, and clinical variables, to improve cancer diagnosis and prognosis modeling.

These advances demonstrate the benefit of carefully designed interaction learning across modalities. However, generic multi-modal fusion frameworks may not sufficiently account for the pronounced heterogeneity in RIL prediction, particularly between anatomical CT representations and spatial dose distributions, which can lead to imbalanced collaboration across modalities. This motivates multi-modal designs that explicitly handle inter-modal heterogeneity and model clinically conditioned cross-modal interactions for effective RIL prediction.

3. Methodology

3.1. Overview

The proposed AANet (Fig. 2) adopts an Align-then-Aggregate strategy to model the complex relationships between diverse data and RIL grades. In the first stage, AANet achieves hierarchical alignment of heterogeneous image features by multi-perspective multi-modal feature fusion, enabling the fused features to be homogeneous and complementary. In the second stage, AANet introduces a clinically guided feature aggregation of aligned image modality information, facilitating the modeling of collaborative interactions between modalities. Aligned features from different modalities are aggregated by concatenation to make the final RIL prediction, which can obtain a comprehensive representation of complex influencing factors and effectively model the impact of inter-modality relationships. The two-stage modeling process can be detailed as follows:

- (1) In the first stage, **Semantic-aligned Attention Multi-modal Fusion (SAMF)** and **Spatial-aligned Multi-perspective Multi-modal Fusion (SMMF)** are designed to hierarchically align features from CT and dose modalities by multi-perspective multi-modal fusion, which can eliminate the heterogeneity within the image modality and between the image and clinical modalities, guiding the image modality features to be complementary across multiple levels. In this stage, SAMF and SMMF shares two 3D-ResNet branches, which extract the CT image features $F_i^C \in \mathbb{R}^{c_i \times h_i \times w_i \times d_i}$ and dose map features $F_i^D \in \mathbb{R}^{c_i \times h_i \times w_i \times d_i}$ respectively, where $i = 1, 2, 3, 4$ denotes different layers, C means CT, and D means dose. By adaptively fusing the CT image features and dose map features at different scales and levels, SAMF and SMMF finally get two multi-modal fusion features E_T and E_S , which exhibit full-scale inter-modal interaction and implicitly achieve preliminary realization of dimensional alignment with metadata.
- (2) In the second stage, **Biased-guided Attention Multi-modal Aggregation (BAMA)** is designed to aggregate the aligned image features E_T and E_S to E_G with the guidance of metadata, which provides the model with clinical context-aware anatomical and quantitative knowledge, thereby enabling more effective modeling of complementary interactions between modalities for RIL

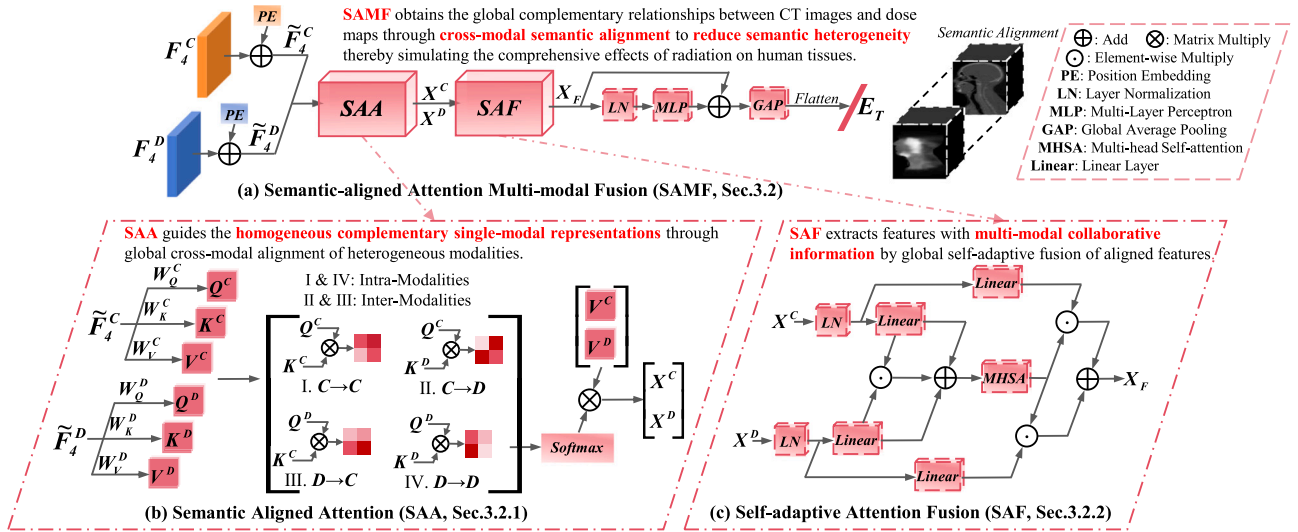


Fig. 3. Architecture of the SAMF module. SAMF performs semantic alignment between CT and dose modalities and consists of two components: SAA guides the complementary homogeneous representations for each modality, and SAF extracts multi-modal collaborative information from the aligned single-modal features.

prediction. Based on the patient's clinical status, BAMA leverages biased cross-modal attention to adaptively focus on the CT and dose features most impactful to lymphocytes, and ultimately obtains inter-modal interaction features E_M . Then, AANet aggregates the above three extracted multi-modal aligned fusion features by concatenation and feeds them into a classification layer to obtain the final RIL prediction results.

3.2. Semantic-aligned Attention Multi-modal Fusion (SAMF)

Semantic-aligned Attention Multi-modal Fusion (SAMF, Fig. 3) is proposed to align CT and dose modalities at the abstract feature level and learn a semantically complementary fused representation, thereby alleviating the interference of inter-modal heterogeneity in pathological representation and achieving preliminary dimensional alignment with patient metadata. SAMF consists of **Semantic Aligned Attention (SAA, Fig. 3(b))** and **Self-adaptive Attention Fusion (SAF, Fig. 3(c))**. Specifically, SAA aligns unimodal semantic information through a multi-branch cross-modal attention mechanism, while SAF further integrates the aligned unimodal features into a multi-modal complementary pathological representation. By adaptively directing attention to informative regions, SAMF preserves cross-modal semantic context within multi-modal representations.

As shown in Fig. 3(a), given the input CT feature $F_4^C \in \mathbb{R}^{C \times H \times W \times D}$ and dose feature $F_4^D \in \mathbb{R}^{C \times H \times W \times D}$ extracted by two symmetric 3D-ResNet branches, we first reshape them into token sequences $X^C \in \mathbb{R}^{N \times C}$ and $X^D \in \mathbb{R}^{N \times C}$, where $N = HWD$. Since this flattening operation removes the explicit spatial arrangement of the original 3D feature maps, modality-specific learnable positional embeddings $P^C \in \mathbb{R}^{N \times C}$ and $P^D \in \mathbb{R}^{N \times C}$ are added to preserve the spatial identity of CT and dose tokens:

$$\tilde{X}^C = X^C + P^C, \quad \tilde{X}^D = X^D + P^D. \quad (1)$$

Here, $\tilde{X}^C$ and $\tilde{X}^D$ denote the position-encoded CT and dose token sequences used as the inputs of SAA. The positional embeddings are not used to directly perform semantic alignment. Instead, they preserve spatial token identity after flattening, which provides the positional basis for the subsequent attention-based semantic alignment in SAA.

The one-to-one correspondence between CT and dose is first established at the data level by paired patient data, spatial registration, and resampling. After preprocessing, CT images and dose maps of the same patient share the same spatial resolution and are processed by

two symmetric 3D-ResNet branches. Therefore, the extracted CT and dose feature maps have the same spatial dimensions, and the n th CT token and the n th dose token correspond to the same spatial location of the same patient after flattening. Based on this deterministic correspondence, SAA further performs classification-oriented semantic alignment through inter- and intra-modal attention. This alignment does not force CT and dose features to become identical representations. Instead, it encourages the two modalities to become more consistent in RIL-grade-related discriminative patterns, so that CT and dose features can collaboratively contribute to the final RIL classification.

3.2.1. Semantic Aligned Attention (SAA)

SAA (Fig. 3(b)) is developed to semantically align image modalities and eliminate abstract inter-modal heterogeneity. Through four directional attention branches, namely CT-to-CT, CT-to-dose, dose-to-CT, and dose-to-dose, it enhances semantic complementarity across modalities and avoids conflicts in high-level modal information during RIL prediction.

For each modality, the flattened token sequence is projected by modality-specific linear mappings to obtain its query, key, and value representations. Specially, three weight matrices are defined for each modality: query weight matrices $W_Q^C, W_Q^D \in \mathbb{R}^{d_s \times d_s}$, key weight matrices $W_K^C, W_K^D \in \mathbb{R}^{d_s \times d_s}$, and value weight matrices $W_V^C, W_V^D \in \mathbb{R}^{d_s \times d_s}$. We compute the query, key, and value vector of each modality, and the Q , K , and V of SAA can be defined as follows:

$$\begin{bmatrix} Q & K & V \end{bmatrix} = \begin{bmatrix} Q^C & K^C & V^C \\ Q^D & K^D & V^D \end{bmatrix} = \begin{bmatrix} W_Q^C \tilde{F}_4^C & W_K^C \tilde{F}_4^C & W_V^C \tilde{F}_4^C \\ W_Q^D \tilde{F}_4^D & W_K^D \tilde{F}_4^D & W_V^D \tilde{F}_4^D \end{bmatrix} \quad (2)$$

where Q^C, K^C, V^C are obtained from $\tilde{F}_4^C$, and Q^D, K^D, V^D are obtained from $\tilde{F}_4^D$. Therefore, we can utilize the MHSA to calculate the long-term relationship within the modalities and obtain the intra- and inter-modal aligned features $X^C \in \mathbb{R}^{d_s \times c_4}$ and $X^D \in \mathbb{R}^{d_s \times c_4}$. X^C and X^D are computed as follows:

$$\begin{bmatrix} X^C \\ X^D \end{bmatrix} = \text{Softmax} \left(\frac{QK^T}{\sqrt{\lambda_1}} \right) V = \begin{bmatrix} A^{C \rightarrow C} & A^{C \rightarrow D} \\ A^{D \rightarrow C} & A^{D \rightarrow D} \end{bmatrix} \begin{bmatrix} V^C \\ V^D \end{bmatrix} \quad (3)$$

where λ_1 is a hyper-parameter. In Eq. (3), the multi-modal attention can be decomposed into four terms pertaining to intra- and inter-modal guidance: (1) the intra-modal CT attention $A^{C \rightarrow C} \in \mathbb{R}^{d_s \times d_s}$, guiding the alignment of CT-to-CT, (2) the inter-modal CT-guided attention

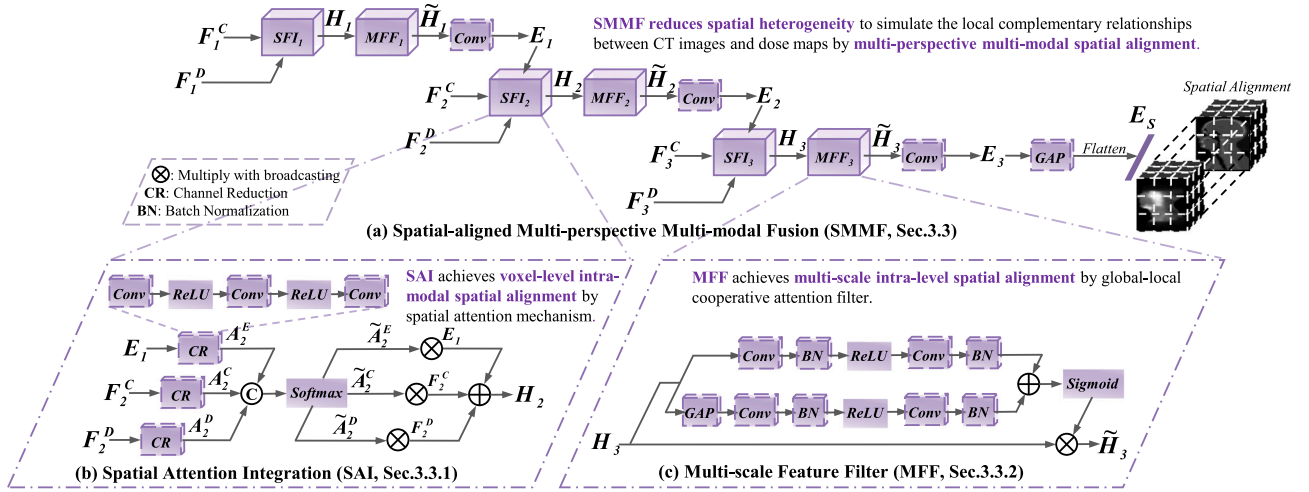


Fig. 4. Architecture of the SMMF module. SMMF performs spatial alignment of CT and dose modalities and consists of two components: SAI aligns features across modalities at the current level, MFF aligns features across depth levels. These components are cascaded to enable multi-perspective multi-modal alignment.

$A^{C \rightarrow D} \in \mathbb{R}^{d_s \times d_s}$, guiding the alignment of CT-to-dose, (3) the inter-modal dose-guided attention $A^{D \rightarrow C} \in \mathbb{R}^{d_s \times d_s}$, guiding the alignment of dose-to-CT, (4) the intra-modal dose attention $A^{D \rightarrow D} \in \mathbb{R}^{d_s \times d_s}$, guiding the alignment of dose-to-dose. Through the guidance of the four-stage interaction process, intra-modal information is transformed into aligned and complementary representations.

3.2.2. Self-adaptive Attention Fusion (SAF)

SAF (Fig. 3(c)) is developed to learn a semantically collaborative representation, which is fused from high-level aligned features. Through self-adaptive multi-modal fusion, SAF further explores collaborative modeling across aligned complementary modalities and achieves preliminary dimensional alignment with metadata.

SAF first mixes the aligned features X^C and X^D output by the SAA to generate the initial hybrid feature $H \in \mathbb{R}^{d_s \times c_4}$. Based on Eq. (4), The process of acquiring H can be expressed by the following equation:

$$H = \text{Linear}(X^C) \odot \text{Linear}(X^D) + X^C + X^D \quad (4)$$

where $\odot$ is the element-wise multiplication operation. The initial hybrid feature is then input into the MHSA layer to capture higher-order dependencies and complementary interactions, resulting in a self-adaptive matrix for multi-modal fusion. The MHSA calculation formula is as follows [27]:

$$\text{MHSA}(H) = \text{Concat}_{m=1}^M \left[\text{Softmax} \left(\frac{Q_m^H (K_m^H)^T}{\sqrt{\lambda_2}} \right) V_m^H \right] W^O, \quad (5)$$

$$Q_m^H = H W_m^Q, \quad K_m^H = H W_m^K, \quad V_m^H = H W_m^V.$$

where M denotes the number of attention heads; Q_m^H , K_m^H , and V_m^H are the query, key, and value representations of the m th head, respectively, obtained from H via the learnable projection matrices W_m^Q , W_m^K , and W_m^V ; λ_2 is the scaling factor used to stabilize the dot-product attention; Softmax normalizes the similarity scores into attention weights; $\text{Concat}_{m=1}^M$ denotes the concatenation of all head outputs along the feature dimension; and W^O is the output projection matrix used to fuse the multi-head information and produce the final self-adaptive fusion representation $\text{MHSA}(H)$, which enhances complementary fusion in a context-aware manner. Therefore, the process of the self-adaptive multi-modal fusion can be expressed by:

$$\begin{aligned} \tilde{X}^C &= \text{MHSA}(H) \odot \text{Linear}(X^C) \\ \tilde{X}^D &= \text{MHSA}(H) \odot \text{Linear}(X^D) \\ X_F &= \tilde{X}^C + \tilde{X}^D \end{aligned} \quad (6)$$

where $X_F \in \mathbb{R}^{d_s \times c_4}$ is the self-adaptive multi-modal fusion feature, which contains abstract multi-modal complementary information.

3.3. Spatial-aligned Multi-perspective Multi-modal Fusion (SMMF)

Spatial-aligned Multi-perspective Multi-modal Fusion (SMMF, Fig. 4) is proposed to align CT and dose modalities at the shallow feature level and learn a spatially complementary fused representation, thereby reducing the effect of inter-modal heterogeneity on spatial correspondence and achieving preliminary dimensional alignment with patient metadata. SMMF contains three pairs of **Spatial Attention Integration (SAI)** and **Multi-scale Feature Filter (MFF)** for three distinct feature levels, where SAI aligns features across modalities by multi-modal fusion, and MFF progressively aligns features across depth levels by multi-scale filtering. Through multi-level cascading, SMMF captures rich spatial information with inter-modal complementarity.

As shown in Fig. 4(a), the overall process of SMMF is as follows:

$$\begin{aligned} \tilde{H}_1 &= \text{Conv}(\text{MFF}_1(\text{SAI}_1(F_1^C, F_1^D))) \\ \tilde{H}_2 &= \text{Conv}(\text{MFF}_2(\text{SAI}_2(F_2^C, F_2^D, \tilde{H}_1))) \\ \tilde{H}_3 &= \text{Conv}(\text{MFF}_3(\text{SAI}_3(F_3^C, F_3^D, \tilde{H}_2))) \\ E_S &= \text{GAP}(\tilde{H}_3) \end{aligned} \quad (7)$$

where SAI_1 , SAI_2 and SAI_3 denote the SAI layers. MFF_1 , MFF_2 and MFF_3 denote the MFF layers. Conv means the kernel size in convolution is $3 \times 3 \times 3$ and the output channel number is twice as many as the input channel number. $E_S \in \mathbb{R}^{c_4}$ is the output of SMMF, that is, the resulting multi-modal aligned feature. By cascading a pair of SAI and MFF modules for each of the three distinct feature levels, SMMF enables fine-grained alignment of shallow features in 3D modalities across multiple scales and hierarchical levels.

3.3.1. Spatial Attention Integration (SAI)

SAI (Fig. 4(b)) is designed to align features from different modalities, learning the spatial correspondence between shallow 3-dimensional modalities, thereby extracting integrated features with enhanced multi-modal perception capabilities for RIL prediction at the same spatial level. Through learnable weighted feature summation, SAI can adaptively adjust the fusion ratio for each voxel, thereby performing voxel-level multi-modal alignment. It is important to note that SAI_1 only accepts two input features because it does not have an upper layer.

In the case of SAI_2 , first, three channel reduction (CR) layers are applied to reduce the channel numbers of F_2^C , F_2^D , and $\tilde{H}_1$ to 1. The CR layer consists of three $1 \times 1 \times 1$ convolutions, which gradually reduce the number of channels by 2 times, 4 times, and finally to 1, and the resulting features are denoted as A_2^C , A_2^D , and A_2^H , respectively. In order to enhance the nonlinearity of the network, a ReLU function

Table 1

Architectural summary of AANet with key module designs.

Module	Main operation	Input	Output	Activation	Role
Input	CT, dose, and metadata	$1 \times 128 \times 128 \times 128$; $1 \times 128 \times 128 \times 128$; 8	–	–	Provides multi-modal patient inputs for end-to-end prediction
Backbone	Two parallel 3D-ResNet18 encoders	CT/dose volume	$F_1: 64 \times 32^3$, F_2 : 128×16^3 , F_3 : 256×8^3 , F_4 : 512×4^3	ReLU	Extracts hierarchical unimodal features for subsequent alignment
SAMF	Flatten F_4 into 64 tokens, add learnable PE, then apply SAA and SAF, followed by FFN and GAP	F_4^C , F_4^D	$E_T \in \mathbb{R}^{512}$	Softmax, GELU	Aligns high-level semantics for complementary multi-modal fusion
SAA	8-head inter-/intra-modal attention with modality-specific Q/K/V projections	$\tilde{F}_4^C$, $\tilde{F}_4^D$	X^C , $X^D \in \mathbb{R}^{64 \times 512}$	Softmax	Aligns CT and dose semantics for reducing abstract inter-modal heterogeneity
SAF	Hybrid fusion $H = \text{Linear}(X^C) \odot \text{Linear}(X^D) + X^C + X^D$, then MHSA and projection	X^C , X^D	$X_F \in \mathbb{R}^{64 \times 512}$	Softmax	Fuses aligned semantic features for collaborative and complementary representation learning
SMMF	Three cascaded SAI-MFF-Conv blocks, where Conv uses kernel $3 \times 3 \times 3$, stride 2, and padding 1 to downsample, followed by GAP	(F_3^C, F_3^D) , (F_2^C, F_2^D) , (F_1^C, F_1^D)	$E_S \in \mathbb{R}^{512}$	ReLU, Softmax, Sigmoid	Aligns shallow spatial structures for progressive multi-level multi-modal fusion
SAI	Three $1 \times 1 \times 1$ convolutions for channel reduction ($C \rightarrow C/2 \rightarrow C/4 \rightarrow 1$), then voxel-wise softmax fusion	Current-level CT/dose/upper-level feature	H_i	ReLU, Softmax	Learns voxel-wise correspondence for same-level multi-modal alignment
MFF	Local branch: Conv-BN-ReLU-Conv-BN; global branch: GAP-Conv-BN-ReLU-Conv-BN; sigmoid-based attention weighting	H_i	$\tilde{H}_i$	ReLU, Sigmoid	Filters cross-level feature conflicts for multi-scale spatial alignment
Meta branch	Two-layer MLP	Metadata: 8	$F_M \in \mathbb{R}^{128}$	GELU	Encodes clinical metadata for interaction guidance
BAMA	Query from F_M ; key/value from $[F_M; E_T; E_S]$; biased cross-modal attention with residual MLP	$[E_T; E_S] \in \mathbb{R}^{1024}$, $F_M \in \mathbb{R}^{128}$	$E_M \in \mathbb{R}^{128}$	Softmax, GELU	Aggregates aligned imaging features with metadata for interaction modeling
Classifier	Fully connected layer	$[E_T; E_S; E_M] \in \mathbb{R}^{1152}$	$N_{\text{class}} = 3$	–	Produces final RIL grade prediction from aggregated representations
Overall complexity: Trainable parameters = 75.97M; FLOPs $\sim$ 65.65G per forward pass (batch size = 1).					

is inserted after the first and second convolution operations of the CR layer. Afterward, A_2^C , A_2^D , and A_2^H are concatenated in the channel dimension to form a three-channel feature. This three-channel feature is processed by a softmax function to generate three voxel-wise attention maps (denoted as $\tilde{A}_2^C$, $\tilde{A}_2^D$, and $\tilde{A}_2^H$):

$$\begin{aligned}\tilde{A}_2^C(j) &= \frac{e^{A_2^C(j)}}{e^{A_2^C(j)} + e^{A_2^D(j)} + e^{A_2^H(j)}} \\ \tilde{A}_2^D(j) &= \frac{e^{A_2^D(j)}}{e^{A_2^C(j)} + e^{A_2^D(j)} + e^{A_2^H(j)}} \\ \tilde{A}_2^H(j) &= \frac{e^{A_2^H(j)}}{e^{A_2^C(j)} + e^{A_2^D(j)} + e^{A_2^H(j)}}\end{aligned}\quad (8)$$

where $A_2^C(j)$, $A_2^D(j)$, $A_2^H(j)$, $\tilde{A}_2^C(j)$, $\tilde{A}_2^D(j)$, and $\tilde{A}_2^H(j)$ represent the elements at the j th position of A^C , A^D , A^H , $\tilde{A}^C$, $\tilde{A}^D$, and $\tilde{A}^H$, respectively. Finally, $\tilde{A}_2^C$, $\tilde{A}_2^D$, and $\tilde{A}_2^H$ are taken as the weights of F_2^C , F_2^D , and $\tilde{H}_1$, and the initial integrated feature H_2 are obtained by weighted summation:

$$H_2 = \tilde{A}_2^C \otimes F_2^C + \tilde{A}_2^D \otimes F_2^D + \tilde{A}_2^H \otimes \tilde{H}_1 \quad (9)$$

where $\otimes$ denotes the element-wise multiplication with broadcasting.

3.3.2. Multi-scale Feature Filter (MFF)

MFF (Fig. 4(c)) is designed to align the initial integrated features H_i at depth level by filtering their conflicts with the next-level features, thereby enabling cross-modal alignment across different levels to progress through stages, allowing the model to acquire multi-level spatial receptive fields. By integrating features at multiple scales, MFF effectively captures both local and global contextual information during

feature fusion and further achieves preliminary dimensional alignment with patient metadata.

MFF_3 obtains the weights of the initial integrated feature H_3 at different positions through a local context unit and a global context unit. Specifically, the local context and the global context are achieved by voxel-wise convolution and global average pooling, respectively. For an input feature $H_3 \in \mathbb{R}^{C \times H \times W \times D}$, the local branch applies two $1 \times 1 \times 1$ convolutions with BN and ReLU to generate a local attention term $A_3^L \in \mathbb{R}^{B \times C \times H \times W \times D}$. In parallel, the global branch first compresses H_3 by global average pooling into a channel descriptor of size $C \times 1 \times 1 \times 1$, and then applies two $1 \times 1 \times 1$ convolutions with BN and ReLU to obtain a global attention term $A_3^G \in \mathbb{R}^{B \times C \times 1 \times 1 \times 1}$. The fusion feature $\tilde{H}_3$ can be obtained by performing a multiplication operation on A_3^S and H_3 , where the global attention term is then broadcast to match the spatial size of the local attention term before element-wise addition, so that local spatial cues and global channel-wise context can be jointly integrated into the final multi-scale attention map. The process can be clearly presented by the following equations:

$$\begin{aligned}A_3^L &= BN(Conv(ReLU(BN(Conv(H_3)))))) \\ A_3^G &= BN(Conv(ReLU(BN(Conv(GAP(H_3)))))) \\ A_3^S &= Sigmoid(A_3^L + A_3^G) \\ \tilde{H}_3 &= A^S \odot H_3\end{aligned}\quad (10)$$

where $Conv$, BN , and $\odot$ denote the $1 \times 1 \times 1$ convolution, batch normalization, and element-wise multiplication, respectively. The $1 \times 1 \times 1$ convolution is employed to implement a linear transformation.

BAMA leverages metadata to guide aligned features by **biased co-guide attention**, thereby modeling inter-modal interactions and incorporating clinical context to assess the impact of radiation.

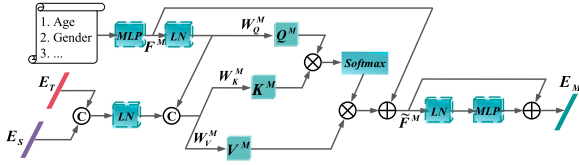


Fig. 5. Architecture of the BAMA module. BAMA aggregates multi-modal information through biased cross-modal attention to model inter-modal interactions.

3.4. Biased-guided Attention Multi-modal Aggregation (BAMA)

BAMA (Fig. 5) is proposed to model collaborative multi-modal interactions by aggregating the aligned semantic and spatial imaging features with metadata, yielding aggregated features that comprehensively integrate clinical, anatomical, and dosimetric information for RIL prediction. Through a meta-guided biased cross-attention mechanism, BAMA adaptively constructs a joint multi-modal representation with clinical context awareness.

BAMA takes the metadata feature F^M obtained by an MLP layer as the query, the concatenation of F^M , E_T , and E_S is used as key and value, then we have [28]:

$$\begin{aligned} Q^M &= W_Q^M \text{LN}(F^M) \\ K^M &= W_K^M \text{Concat}(\text{LN}(F^M), \text{LN}(\text{Concat}(E_T, E_S))) \\ V^M &= W_V^M \text{Concat}(\text{LN}(F^M), \text{LN}(\text{Concat}(E_T, E_S))) \end{aligned} \quad (11)$$

where Q^M , K^M , and V^M have the same dimensions. Then, leveraging the MHSA mechanism, we can get the output $\tilde{F}^M$ based on Q^M , K^M , V^M from Eq. (11), and feed the output into an MLP layer with a residual shortcut. Finally, we get the fusion feature E_M . The process can be formulated as:

$$\begin{aligned} \tilde{F}^M &= \text{Softmax}\left(\frac{Q^M K^{M^T}}{\sqrt{\lambda_2}}\right) V^M + F^M \\ E_M &= \text{MLP}(\text{LN}(\tilde{F}^M)) + \tilde{F}^M \end{aligned} \quad (12)$$

where λ_2 is a hyper-parameter.

Finally, the outputs from SAMF, SMMF, and BAMA are concatenated and fed into the classification head to predict the patient's RIL grade. To make the overall architecture easier to follow, Table 1 provides a structured summary of AANet, including the main operation, input and output dimensions, activation function, and structural role of each major component and proposed block. Overall, AANet first extracts hierarchical CT and dose features using two symmetric 3D-ResNet branches, then performs high-level semantic alignment through SAMF and multi-level spatial alignment through SMMF, and finally aggregates the aligned imaging features with metadata through BAMA for RIL prediction.

4. Experiments and results

4.1. Dataset

This study encompassed 211 NPC patients, including 40 individuals with grade 2 radiation-induced lymphopenia (G2 RIL), 114 individuals with grade 3 radiation-induced lymphopenia (G3 RIL), and 57 individuals with grade 4 radiation-induced lymphopenia (G4 RIL). Since the majority of NPC patients will experience severe lymphopenia after radiation therapy, the number of NPC patients with grade 1 radiation-induced lymphopenia (G1RIL) is extremely limited. Consequently, patients with G1RIL were not included in this study. All patients were re-staged according to the eighth edition of the International Union Against Cancer/American Joint Committee on Cancer

Table 2

Cohort information of all patients in this study.

	G2 RIL	G3 RIL	G4 RIL
Age(mean ± std)	59 ± 7	51 ± 9	48 ± 10
Gender (%)			
Male	26(12.32%)	84(39.81%)	42(19.91%)
Female	14(6.63%)	30(14.22%)	15(7.11%)
T-stage ^a , No. (%)			
I	2(0.95%)	13(6.16%)	8(3.79%)
II	12(5.69%)	42(19.91%)	9(4.27%)
III	15(7.11%)	44(20.85%)	19(9.00%)
IV	11(5.21%)	15(7.11%)	21(9.95%)
N-stage ^a , No. (%)			
I	2(0.95%)	8(3.79%)	0(0%)
II	13(6.16%)	16(7.58%)	13(6.16%)
III	15(7.11%)	68(32.33%)	28(13.27%)
IV	10(4.74%)	22(10.43%)	16(7.58%)
Overall-stage ^a , No. (%)			
I	0(0%)	1(0.47%)	0(0%)
II	4(1.90%)	14(6.64%)	2(0.95%)
III	16(7.58%)	64(30.33%)	23(10.90%)
IV	20(9.48%)	35(16.58%)	32(15.57%)
BLC ^b (mean ± std)	2.01 ± 0.75	1.71 ± 0.60	1.29 ± 0.55
BNC ^c (mean ± std)	3.80 ± 1.73	3.89 ± 2.00	4.59 ± 2.59

^a According to the 8th edition of the International Union against Cancer/American Joint Committee on Cancer (UICC/AJCC) staging manual.

^b Baseline lymphocyte count.

^c Baseline neutrophil count.

(UICC/AJCC) staging system. Metadata comprises gender, age, tumor stage, baseline lymphocyte count, and baseline neutrophil count, as shown in Table 2.

4.2. Data preprocessing

For CT images and dose maps, we first align them spatially and remove the excess parts from the CT images and dose maps. We then used resampling to sample both the CT image and the dose maps to the size of [128,128,128]. Finally, we used the Z-score method to normalize the CT images and dose maps to avoid the impact of different feature dimensions on the model. For metadata, we used the Z-score method to normalize the continuous data and transformed the discrete data using one-hot encoding. Patient metadata after preprocessing includes gender, age, tumor stage, baseline lymphocyte count, and baseline neutrophil count, which is encoded as an 8-bit vector.

4.3. Evaluation criterion

To better demonstrate the ability of the proposed method to predict RIL grade in NPC, all experimental results were determined by 5-fold cross-validation, and we calculated the area under the curve (AUC), accuracy (ACC) = $\frac{TP+TN}{TP+FP+FN+TN}$, precision (PRE) = $\frac{TP}{TP+FP}$, sensitivity (SEN) = $\frac{TP}{TP+FN}$, specificity (SPE) = $\frac{TN}{TN+FP}$, and F1-score (F1) = $\frac{2 \times \text{SEN} \times \text{PRE}}{\text{SEN} + \text{PRE}}$ to measure the performance of the proposed framework and other methods, where TP , FP , TN , and FN are the numbers of true positive, true negative, false positive, and false negative, respectively.

4.4. Implementation details

The proposed AANet was implemented in PyTorch and trained on a single NVIDIA GeForce RTX 3090 GPU. For feature extraction, the 3D-ResNet18 was used for image encoding, while the metadata branch was implemented using two fully connected layers with GELU activation. The model was trained with the cross-entropy loss under a stratified 5-fold patient-level cross-validation protocol, where patients were allocated to the five folds by proportionally stratifying the RIL grades. All data belonging to the same patient, including CT images,

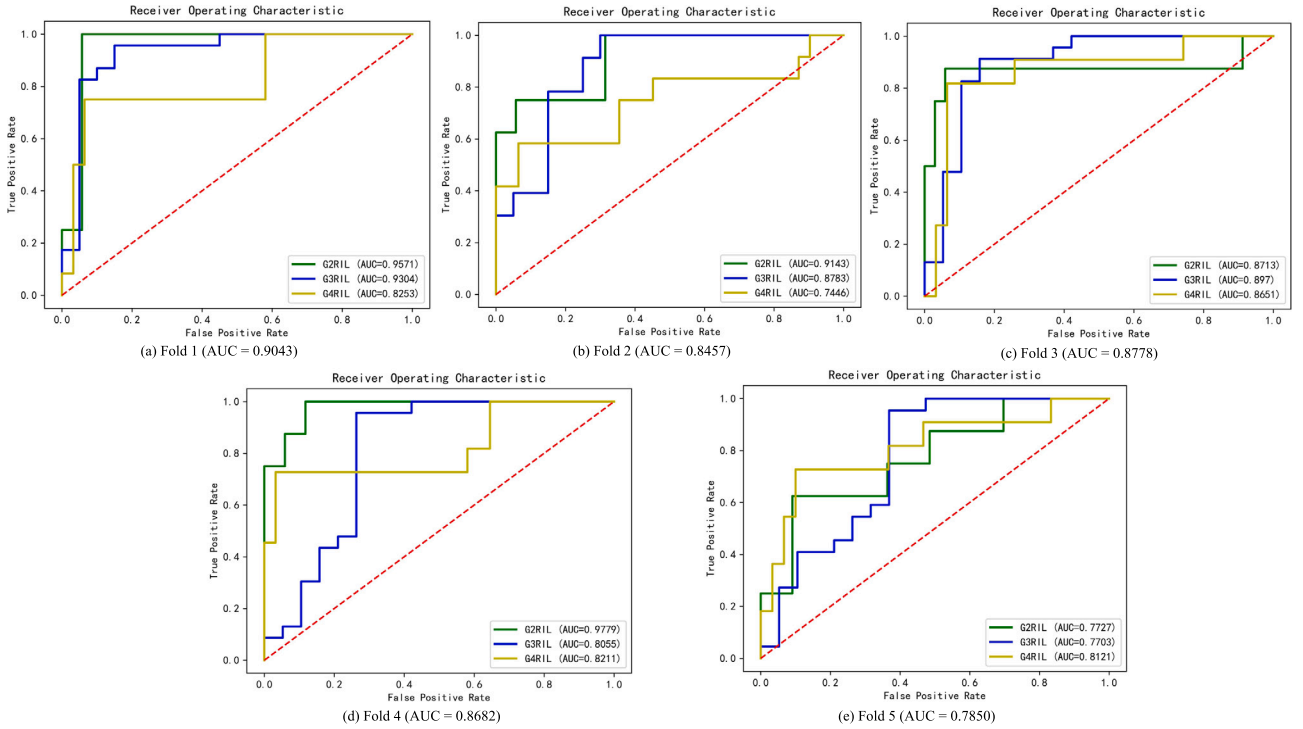


Fig. 6. ROC curves of each fold of the 5-fold cross-validation. The average ROC curves are relatively consistent across folds, indicating that the model shows limited variation in overall performance. However, the individual class performance varies across folds, primarily due to the imbalanced class distribution.

Table 3
Selected training hyperparameters for AANet.

Hyperparameter	Value
Epochs	100
Batch size	4
Optimizer	Adam
Adam β_1	0.9
Initial learning rate	1×10^{-4}
Learning rate decay	Cosine annealing
Weight decay	1×10^{-5}
Dropout	0.2
Random seed	3407

dose maps, metadata, and labels, were assigned to the same fold. In each round, four folds were used for training and the remaining fold was held out exclusively for testing. The held-out fold was never used for model fitting, fold-specific hyperparameter tuning, or any other optimization step, thereby preventing potential data leakage. Standard deviations (SD) are also reported. Selected training hyperparameters are summarized in Table 3.

4.5. Five-fold cross-validation results

Table 4 presents the 5-fold cross-validation results. AANet achieved an AUC of 0.8562 ± 0.0450 , Accuracy of 0.8148 ± 0.0380 , Precision of 0.8375 ± 0.0183 , Sensitivity of 0.7670 ± 0.0808 , Specificity of 0.8870 ± 0.0304 , and F1-score of 0.7784 ± 0.0724 . The relatively small standard deviations across all metrics demonstrate the model's consistent class-average performance across folds. Fig. 6 shows the receiver operating characteristic (ROC) of each fold of the 5-fold cross-validation, which also supports the conclusions drawn above.

4.6. Effect of multi-modal learning in RIL prediction

To evaluate the effect of different modality combinations on RIL prediction, experiments are conducted using single-modality and multi-modality inputs. For multi-modal settings, AANet is compared with two

fusion baselines, namely Concat and MLP. Specifically, Concat means only using concatenation for fusion, and MLP means using a multi-layer perceptron for fusion. Table 5 shows that the best performance is achieved when all modalities are used jointly, whereas removing any modality leads to a clear drop in predictive performance. In addition, under dual-modality settings, both simple fusion and AANet improve over single-modality inputs. However, when all three modalities are used, AANet continues to improve, while the simple Concat baseline does not show further gains.

4.7. Ablation study

To evaluate the contribution of each design in AANet, a series of ablation studies are conducted on the main modules and their internal components. As shown in Table 6, the complete AANet achieves the best overall performance. When only SAMF is introduced, the model obtains an AUC of 0.7860 ± 0.0231 and an ACC of 0.7535 ± 0.0122 ; using only SMMF yields an AUC of 0.7692 ± 0.0385 and an ACC of 0.7346 ± 0.0088 ; and using only BAMA yields an AUC of 0.7362 ± 0.0621 and an ACC of 0.7390 ± 0.0357 . In contrast, the simple Concat baseline achieves only 0.6966 ± 0.0411 in AUC and 0.6966 ± 0.0265 in ACC, while the MLP baseline reaches 0.7588 ± 0.0863 in AUC and 0.7348 ± 0.0366 in ACC. These results show that all three main modules contribute to the final performance, and their combination gives the strongest prediction results.

Tables 8–10 further analyze the internal designs of the proposed modules. For SAMF, the ablation results in Table 8 demonstrate the positive impact of alignment attention across various dimensions on semantic alignment of SAA, enabling effective prediction outcomes. Specifically, the model achieved optimal performance when the SAA simultaneously aligned different modal semantics from four directions. Besides, the attention ablation in Table 7 demonstrates the effectiveness of MHSA. Specifically, MHSA achieves the best overall performance compared to different attention designs, including the simple linear layer and the single-head self-attention (SHSA). This indicates that the effectiveness of SAF is not merely due to an additional projection or

Table 4

Five-fold cross-validation performances demonstrate AANet's generalization capability and robustness.

Fold	AUC(↑)	ACC(↑)	PRE(↑)	SEN(↑)	SPE(↑)	F1(↑)
1	0.9043	0.8372	0.8121	0.8454	0.9166	0.8194
2	0.8457	0.8140	0.8570	0.7361	0.8785	0.7735
3	0.8778	0.8571	0.8501	0.8271	0.9161	0.8373
4	0.8682	0.8095	0.8424	0.7823	0.8800	0.8059
5	0.7850	0.7561	0.8258	0.6439	0.8439	0.6558
mean	0.8562	0.8148	0.8375	0.7670	0.8870	0.7784
std	0.0450	0.0380	0.0183	0.0808	0.0304	0.0724

Table 5

Modality ablation studies demonstrate that integrating additional modalities enhances model performance, and our proposed Align-then-Aggregate fusion strategy surpasses existing approaches.

	CT	Dose	Meta	AUC(↑)	ACC(↑)	PRE(↑)	SEN(↑)	SPE(↑)	F1(↑)
Single-modality	✓			0.6936 ± 0.0227	0.6921 ± 0.0316	0.6109 ± 0.1287	0.5543 ± 0.0351	0.8125 ± 0.0237	0.5375 ± 0.0435
		✓		0.6577 ± 0.0236	0.6824 ± 0.0124	0.5263 ± 0.1284	0.5290 ± 0.0377	0.7958 ± 0.0129	0.5077 ± 0.0680
			✓	0.6549 ± 0.0432	0.6161 ± 0.0418	0.6177 ± 0.0856	0.5362 ± 0.0528	0.7598 ± 0.0260	0.5519 ± 0.0651
Concat	✓	✓		0.6745 ± 0.0505	0.7156 ± 0.0242	0.7203 ± 0.0882	0.5894 ± 0.0249	0.8275 ± 0.0163	0.5797 ± 0.0486
	✓		✓	0.7131 ± 0.0356	0.7154 ± 0.0197	0.6491 ± 0.0926	0.5945 ± 0.0188	0.8390 ± 0.0073	0.5780 ± 0.0253
		✓	✓	0.6945 ± 0.0226	0.6777 ± 0.0213	0.5337 ± 0.1290	0.5240 ± 0.0519	0.7899 ± 0.0239	0.5050 ± 0.0810
MLP	✓	✓		0.6966 ± 0.0411	0.6966 ± 0.0265	0.5937 ± 0.0927	0.5620 ± 0.0316	0.8146 ± 0.0178	0.5537 ± 0.0521
	✓		✓	0.7362 ± 0.0660	0.6968 ± 0.0371	0.6263 ± 0.1142	0.6086 ± 0.0663	0.8227 ± 0.0268	0.6058 ± 0.0881
	✓	✓	✓	0.7377 ± 0.0338	0.7441 ± 0.0073	0.6155 ± 0.1274	0.6365 ± 0.0295	0.8538 ± 0.0072	0.6105 ± 0.0734
Ours		✓	✓	0.7464 ± 0.0566	0.7157 ± 0.0328	0.7502 ± 0.0747	0.6222 ± 0.0598	0.8211 ± 0.0276	0.6365 ± 0.0611
	✓	✓	✓	0.7588 ± 0.0863	0.7348 ± 0.0366	0.6826 ± 0.1413	0.6451 ± 0.0680	0.8481 ± 0.0235	0.6358 ± 0.0939
	✓	✓		0.7048 ± 0.0238	0.7302 ± 0.0220	0.7133 ± 0.0829	0.5986 ± 0.0268	0.8381 ± 0.0180	0.5990 ± 0.0338
Ours	✓		✓	0.7631 ± 0.0558	0.7490 ± 0.0376	0.7083 ± 0.0682	0.6413 ± 0.0596	0.8497 ± 0.0228	0.6530 ± 0.0666
		✓	✓	0.7725 ± 0.0469	0.7302 ± 0.0277	0.7440 ± 0.0446	0.6300 ± 0.0566	0.8320 ± 0.0248	0.6437 ± 0.0467
	✓	✓	✓	0.8562 ± 0.0450	0.8148 ± 0.0380	0.8375 ± 0.0183	0.7670 ± 0.0808	0.8870 ± 0.0304	0.7784 ± 0.0724

CT, Dose, and Meta means CT image modality, dose map modality, and metadata modality, respectively.

Concat, MLP, and AANet are all multi-modal learning methods.

Table 6

Ablation studies show the effectiveness of each component in our AANet for RIL prediction.

Concat	MLP	SAMF		SMMF		BAMA	AUC(↑)	ACC(↑)	PRE(↑)	SEN(↑)	SPE(↑)	F1(↑)
		SAA	SAF	SAI	MFF							
✓							0.6966 ± 0.0411	0.6966 ± 0.0265	0.5937 ± 0.0927	0.5620 ± 0.0316	0.8146 ± 0.0178	0.5537 ± 0.0521
	✓						0.7588 ± 0.0863	0.7348 ± 0.0366	0.6826 ± 0.1413	0.6451 ± 0.0680	0.8481 ± 0.0235	0.6358 ± 0.0939
		✓	✓				0.7860 ± 0.0231	0.7535 ± 0.0122	0.7766 ± 0.0692	0.6580 ± 0.0449	0.8463 ± 0.0132	0.6714 ± 0.0490
				✓	✓		0.7692 ± 0.0385	0.7346 ± 0.0088	0.7520 ± 0.0457	0.6125 ± 0.0286	0.8268 ± 0.0102	0.6291 ± 0.0379
						✓	0.7362 ± 0.0621	0.7390 ± 0.0357	0.7504 ± 0.0815	0.6459 ± 0.0559	0.8393 ± 0.0242	0.6594 ± 0.0481
		✓	✓	✓	✓		0.7791 ± 0.0456	0.7726 ± 0.0257	0.7682 ± 0.0448	0.6726 ± 0.0264	0.8598 ± 0.0216	0.6900 ± 0.0218
		✓	✓			✓	0.8227 ± 0.0424	0.7862 ± 0.0410	0.8173 ± 0.0661	0.7189 ± 0.0752	0.8627 ± 0.0337	0.7397 ± 0.0588
				✓	✓	✓	0.8149 ± 0.0361	0.7486 ± 0.0331	0.7165 ± 0.0587	0.6859 ± 0.0686	0.8540 ± 0.0181	0.6879 ± 0.0661
		✓		✓	✓	✓	0.8227 ± 0.0555	0.7678 ± 0.0340	0.7806 ± 0.0731	0.6906 ± 0.0478	0.8568 ± 0.0235	0.7111 ± 0.0434
			✓	✓	✓	✓	0.8333 ± 0.0527	0.7536 ± 0.0348	0.7781 ± 0.0597	0.6706 ± 0.0483	0.8465 ± 0.0214	0.6909 ± 0.0470
		✓	✓	✓		✓	0.8214 ± 0.0306	0.7816 ± 0.0346	0.8107 ± 0.0826	0.7036 ± 0.0577	0.8622 ± 0.0207	0.7261 ± 0.0650
		✓	✓		✓	✓	0.7516 ± 0.0715	0.7015 ± 0.0631	0.6687 ± 0.1120	0.6100 ± 0.0750	0.8306 ± 0.0367	0.6101 ± 0.0749
	✓	✓	✓	✓	✓	✓	0.8562 ± 0.0450	0.8148 ± 0.0380	0.8375 ± 0.0183	0.7670 ± 0.0808	0.8870 ± 0.0304	0.7784 ± 0.0724

Table 7

Attention ablation in SAF demonstrates the effectiveness of MHSA.

Method	Stat	AUC(↑)	ACC(↑)	PRE(↑)	SEN(↑)	SPE(↑)	F1(↑)
Linear	mean	0.8427	0.7818	0.8069	0.6990	0.8642	0.7142
	std	0.0705	0.0321	0.0279	0.0774	0.0283	0.0567
SHSA	mean	0.8460	0.7770	0.8008	0.6907	0.8580	0.7107
	std	0.0457	0.0441	0.0566	0.0748	0.0345	0.0731
MHSA	mean	0.8562	0.8148	0.8375	0.7670	0.8870	0.7784
	std	0.0450	0.0380	0.0183	0.0808	0.0304	0.0724

attention-like operation, but is closely related to the ability of MHSA to model complementary cross-modal interactions under global context in multiple representation subspaces.

For SMMF, the level-wise ablation in Table 9 shows that progressively introducing spatial alignment across multiple levels improves the model performance. Specifically, the setting without hierarchical

level selection obtains an AUC of 0.8227 ± 0.0424 and an ACC of 0.7862 ± 0.0410 , while using the first level only improves the AUC to 0.8316 ± 0.0334 and the ACC to 0.7911 ± 0.0344 . When the first two levels are used, the AUC further increases to 0.8390 ± 0.0318 , and the full three-level setting achieves the best performance with an AUC of 0.8562 ± 0.0450 and an ACC of 0.8148 ± 0.0380 .

In addition, Table 10 reports the effect of different contextual filter scales in MFF. Using only local filtering yields an AUC of 0.8303 ± 0.0389 and an ACC of 0.7724 ± 0.0282 , while using only global filtering achieves an AUC of 0.8438 ± 0.0450 and an ACC of 0.7767 ± 0.0405 . The homogeneous combinations Local + Local and Global + Global obtain AUCs of 0.8098 ± 0.0535 and 0.8254 ± 0.0410 , respectively. Among all settings, the Local + Global combination achieves the best performance, with an AUC of 0.8562 ± 0.0450 , an ACC of 0.8148 ± 0.0380 , and an F1 score of 0.7784 ± 0.0724 . Overall, these ablation results support the effectiveness of the full Align-then-Aggregate design in AANet.

Table 8

Direction ablation studies of semantic alignment attention in SAA demonstrate the effectiveness of 4-directional inter- and intra-modal alignment.

$C \rightarrow C$	$D \rightarrow D$	$C \rightarrow D$	$D \rightarrow C$	AUC($\uparrow$)	ACC($\uparrow$)	PRE($\uparrow$)	SEN($\uparrow$)	SPE($\uparrow$)	F1($\uparrow$)
				0.8333 $\pm$ 0.0527	0.7536 $\pm$ 0.0348	0.7781 $\pm$ 0.0597	0.6706 $\pm$ 0.0483	0.8465 $\pm$ 0.0214	0.6909 $\pm$ 0.0470
✓				0.8155 $\pm$ 0.0609	0.7721 $\pm$ 0.0256	0.7856 $\pm$ 0.0668	0.6997 $\pm$ 0.0552	0.8598 $\pm$ 0.0267	0.7194 $\pm$ 0.0448
	✓			0.8399 $\pm$ 0.0253	0.7727 $\pm$ 0.0244	0.7607 $\pm$ 0.0545	0.7133 $\pm$ 0.0422	0.8680 $\pm$ 0.0161	0.7236 $\pm$ 0.0326
		✓		0.8340 $\pm$ 0.0238	0.7814 $\pm$ 0.0390	0.7833 $\pm$ 0.0803	0.7305 $\pm$ 0.0653	0.8689 $\pm$ 0.0290	0.7423 $\pm$ 0.0500
			✓	0.7935 $\pm$ 0.0603	0.7488 $\pm$ 0.0210	0.7610 $\pm$ 0.0526	0.6695 $\pm$ 0.0776	0.8517 $\pm$ 0.0331	0.6738 $\pm$ 0.0584
✓	✓			0.8382 $\pm$ 0.0497	0.7720 $\pm$ 0.0425	0.7608 $\pm$ 0.0655	0.6923 $\pm$ 0.0706	0.8649 $\pm$ 0.0330	0.7048 $\pm$ 0.0685
✓		✓		0.8381 $\pm$ 0.0422	0.7628 $\pm$ 0.0196	0.7554 $\pm$ 0.0645	0.7003 $\pm$ 0.0501	0.8597 $\pm$ 0.0190	0.7064 $\pm$ 0.0481
✓			✓	0.8178 $\pm$ 0.0433	0.7721 $\pm$ 0.0256	0.7992 $\pm$ 0.0736	0.6821 $\pm$ 0.0416	0.8532 $\pm$ 0.0183	0.7099 $\pm$ 0.0441
	✓	✓		0.8345 $\pm$ 0.0375	0.7673 $\pm$ 0.0356	0.7829 $\pm$ 0.0940	0.6959 $\pm$ 0.0730	0.8572 $\pm$ 0.0263	0.7171 $\pm$ 0.0672
	✓		✓	0.8165 $\pm$ 0.0234	0.7628 $\pm$ 0.0196	0.7759 $\pm$ 0.0578	0.6864 $\pm$ 0.0563	0.8567 $\pm$ 0.0159	0.6911 $\pm$ 0.0659
		✓	✓	0.8397 $\pm$ 0.0347	0.7770 $\pm$ 0.0235	0.7814 $\pm$ 0.0509	0.7134 $\pm$ 0.0717	0.8688 $\pm$ 0.0256	0.7208 $\pm$ 0.0586
✓	✓	✓		0.8387 $\pm$ 0.0414	0.7676 $\pm$ 0.0135	0.7670 $\pm$ 0.0480	0.7330 $\pm$ 0.0605	0.8693 $\pm$ 0.0126	0.7305 $\pm$ 0.0383
✓	✓		✓	0.8409 $\pm$ 0.0401	0.7721 $\pm$ 0.0392	0.8108 $\pm$ 0.0425	0.7005 $\pm$ 0.0755	0.8561 $\pm$ 0.0284	0.7151 $\pm$ 0.0814
✓		✓	✓	0.8220 $\pm$ 0.0433	0.7678 $\pm$ 0.0076	0.7925 $\pm$ 0.0571	0.7067 $\pm$ 0.0450	0.8565 $\pm$ 0.0231	0.7141 $\pm$ 0.0119
	✓	✓	✓	0.8081 $\pm$ 0.0325	0.7769 $\pm$ 0.0247	0.7518 $\pm$ 0.0655	0.7162 $\pm$ 0.0490	0.8708 $\pm$ 0.0129	0.7240 $\pm$ 0.0506
✓	✓	✓	✓	0.8562 $\pm$ 0.0450	0.8148 $\pm$ 0.0380	0.8375 $\pm$ 0.0183	0.7670 $\pm$ 0.0808	0.8870 $\pm$ 0.0304	0.7784 $\pm$ 0.0724

Table 9

Level ablation studies of spatial alignment in SMMF show the effectiveness of hierarchical layer-wise alignment at three levels.

F_1^C, F_1^D	F_2^C, F_2^D	F_3^C, F_3^D	AUC($\uparrow$)	ACC($\uparrow$)	PRE($\uparrow$)	SEN($\uparrow$)	SPE($\uparrow$)	F1($\uparrow$)
			0.8227 $\pm$ 0.0424	0.7862 $\pm$ 0.0410	0.8173 $\pm$ 0.0661	0.7189 $\pm$ 0.0752	0.8627 $\pm$ 0.0337	0.7397 $\pm$ 0.0588
		✓	0.8316 $\pm$ 0.0334	0.7911 $\pm$ 0.0344	0.8128 $\pm$ 0.0201	0.7143 $\pm$ 0.0714	0.8692 $\pm$ 0.0266	0.7299 $\pm$ 0.0656
	✓	✓	0.8390 $\pm$ 0.0318	0.7721 $\pm$ 0.0304	0.8262 $\pm$ 0.0453	0.6952 $\pm$ 0.0684	0.8552 $\pm$ 0.0250	0.7194 $\pm$ 0.0633
✓	✓	✓	0.8562 $\pm$ 0.0450	0.8148 $\pm$ 0.0380	0.8375 $\pm$ 0.0183	0.7670 $\pm$ 0.0808	0.8870 $\pm$ 0.0304	0.7784 $\pm$ 0.0724

Table 10

Contextual filter scale ablation studies in MFF demonstrate the effectiveness of integrating global and local perspectives.

Fusion scale	AUC($\uparrow$)	ACC($\uparrow$)	PRE($\uparrow$)	SEN($\uparrow$)	SPE($\uparrow$)	F1($\uparrow$)
Local	0.8303 $\pm$ 0.0389	0.7724 $\pm$ 0.0282	0.7728 $\pm$ 0.0598	0.7233 $\pm$ 0.0588	0.8690 $\pm$ 0.0166	0.7219 $\pm$ 0.0526
Global	0.8438 $\pm$ 0.0450	0.7767 $\pm$ 0.0405	0.7998 $\pm$ 0.0430	0.7065 $\pm$ 0.0864	0.8603 $\pm$ 0.0356	0.7139 $\pm$ 0.0896
Local + Local	0.8098 $\pm$ 0.0535	0.7818 $\pm$ 0.0223	0.8010 $\pm$ 0.0592	0.7162 $\pm$ 0.0546	0.8634 $\pm$ 0.0240	0.7340 $\pm$ 0.0396
Global + Global	0.8254 $\pm$ 0.0410	0.7675 $\pm$ 0.0234	0.7831 $\pm$ 0.0812	0.6906 $\pm$ 0.0577	0.8554 $\pm$ 0.0199	0.7099 $\pm$ 0.0510
Local + Global	0.8562 $\pm$ 0.0450	0.8148 $\pm$ 0.0380	0.8375 $\pm$ 0.0183	0.7670 $\pm$ 0.0808	0.8870 $\pm$ 0.0304	0.7784 $\pm$ 0.0724

Table 11

Comparison experiments demonstrate the effectiveness of the proposed AANet for multi-modal learning in RIL prediction.

Method		AUC($\uparrow$)	ACC($\uparrow$)	PRE($\uparrow$)	SEN($\uparrow$)	SPE($\uparrow$)	F1($\uparrow$)
Fusion stage 1	Fusion stage 2						
CentralNet [29]	Concat	0.6797 $\pm$ 0.0504	0.7154 $\pm$ 0.0197	0.6384 $\pm$ 0.1234	0.5736 $\pm$ 0.0289	0.8256 $\pm$ 0.0137	0.5606 $\pm$ 0.0453
	MLP	0.7630 $\pm$ 0.0335	0.7203 $\pm$ 0.0389	0.6702 $\pm$ 0.0639	0.6112 $\pm$ 0.0512	0.8386 $\pm$ 0.0284	0.6113 $\pm$ 0.0419
MMTM [30]	Concat	0.6976 $\pm$ 0.0450	0.7156 $\pm$ 0.0174	0.5917 $\pm$ 0.0831	0.5909 $\pm$ 0.0595	0.8330 $\pm$ 0.0235	0.5694 $\pm$ 0.0640
	MLP	0.7763 $\pm$ 0.0273	0.7347 $\pm$ 0.0416	0.6885 $\pm$ 0.1541	0.6400 $\pm$ 0.0651	0.8419 $\pm$ 0.0197	0.6376 $\pm$ 0.0985
HcCNN [15]	Concat	0.6957 $\pm$ 0.0658	0.7065 $\pm$ 0.0324	0.6471 $\pm$ 0.1065	0.5872 $\pm$ 0.0378	0.8280 $\pm$ 0.0178	0.5853 $\pm$ 0.0452
	MLP	0.7380 $\pm$ 0.0227	0.7631 $\pm$ 0.0153	0.8094 $\pm$ 0.0369	0.6461 $\pm$ 0.0283	0.8467 $\pm$ 0.0166	0.6693 $\pm$ 0.0286
Yang et al. [16]	Concat	0.6926 $\pm$ 0.0466	0.7346 $\pm$ 0.0300	0.7179 $\pm$ 0.1646	0.6105 $\pm$ 0.0490	0.8273 $\pm$ 0.0161	0.6214 $\pm$ 0.0803
	MLP	0.7275 $\pm$ 0.0266	0.7347 $\pm$ 0.0236	0.7016 $\pm$ 0.1496	0.6140 $\pm$ 0.0550	0.8365 $\pm$ 0.0216	0.6111 $\pm$ 0.0808
MDL-Net [31]	Concat	0.7549 $\pm$ 0.0380	0.7252 $\pm$ 0.0095	0.7515 $\pm$ 0.0777	0.6200 $\pm$ 0.0276	0.8306 $\pm$ 0.0147	0.6311 $\pm$ 0.0159
	MLP	0.7503 $\pm$ 0.0264	0.7394 $\pm$ 0.0226	0.7531 $\pm$ 0.0749	0.6200 $\pm$ 0.0301	0.8339 $\pm$ 0.0183	0.6336 $\pm$ 0.0365
Liu et al. [32]	Liu et al. [32]	0.7823 $\pm$ 0.0428	0.7395 $\pm$ 0.0211	0.7274 $\pm$ 0.0520	0.6418 $\pm$ 0.0693	0.8431 $\pm$ 0.0262	0.6483 $\pm$ 0.0710
Ours		0.8562 $\pm$ 0.0450	0.8148 $\pm$ 0.0380	0.8375 $\pm$ 0.0183	0.7670 $\pm$ 0.0808	0.8870 $\pm$ 0.0304	0.7784 $\pm$ 0.0724

4.8. Comparison with other fusion methods

To validate the superiority of the proposed method in multi-modal fusion RIL prediction, AANet is further compared with several existing multi-modal fusion methods, including CentralNet [29], MMTM [30], HcCNN [15], Yang et al. [16], MDL-Net [31], and Liu et al. [32]. Since some methods are only applicable to the same type of modality, we uniformly perform fusion in two stages for fair comparison, in which metadata is introduced using Concat and MLP, respectively. As shown in Table 11, AANet achieves better overall performance than the methods compared on the main evaluation metrics. Specifically, AANet obtains an AUC of 0.8562 ± 0.0450 , an ACC of 0.8148 ± 0.0380 ,

and an F1 score of 0.7784 ± 0.0724 , which are higher than the best competing results of 0.7823 ± 0.0428 in AUC, 0.7631 ± 0.0153 in ACC, and 0.6693 ± 0.0286 in F1. These results indicate that AANet provides more competitive performance than existing fusion methods in the RIL prediction task.

Furthermore, the radar charts in Fig. 7 visualize the comparison results of Table 11, where the 6 dimensions correspond to the 6 metrics used for comparison. In such plots, a larger enclosed area indicates superior overall performance. As illustrated, AANet consistently occupies the outermost region across all metrics, demonstrating its effectiveness compared to other fusion strategies.

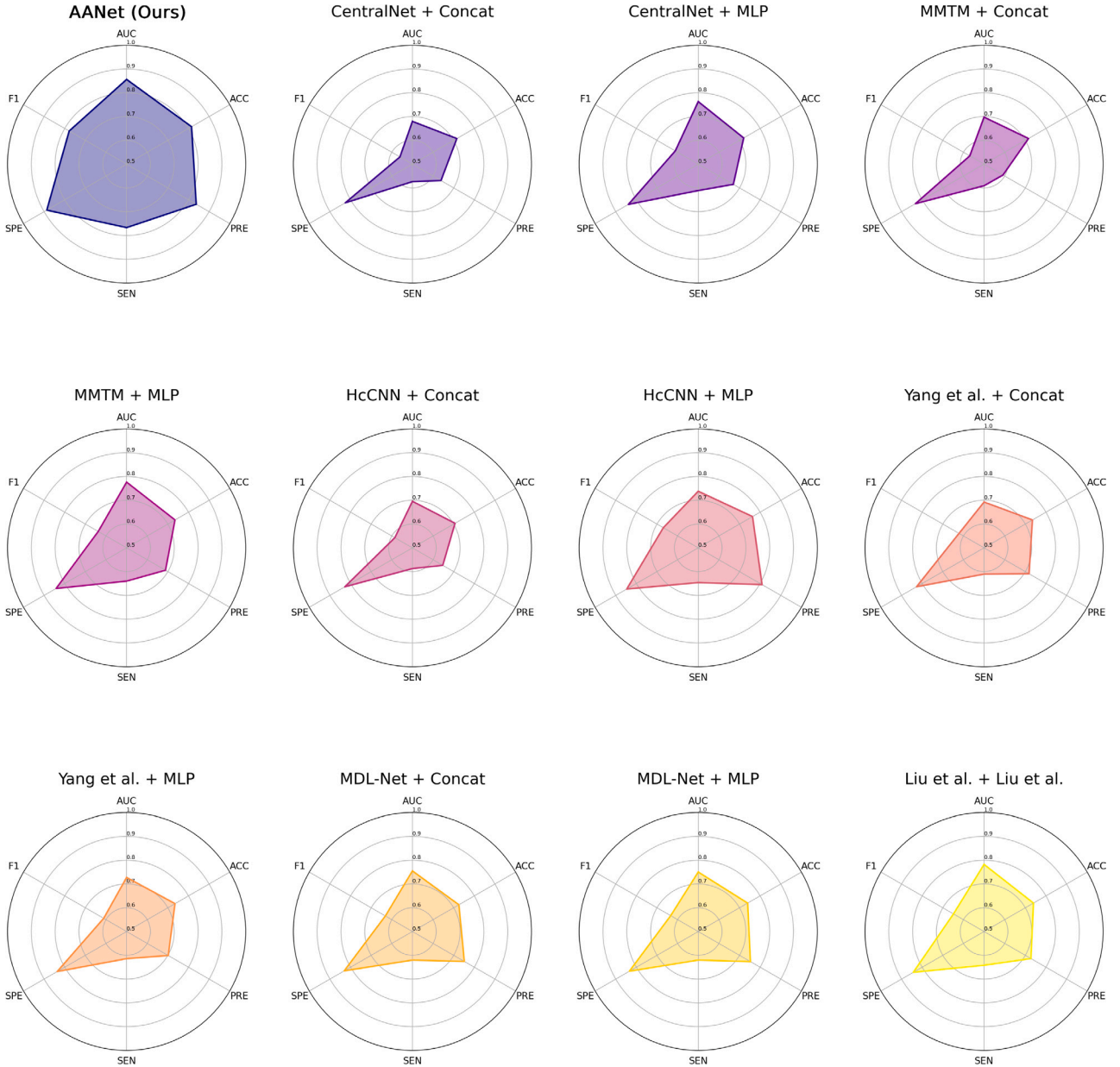


Fig. 7. Radar charts based on the visualization of comparison results, where the 6 dimensions correspond to the 6 metrics of evaluation. The larger radar coverage area of AANet indicates that its overall performance surpasses that of existing methods.

5. Discussion

This study proposes AANet, an end-to-end multi-modal collaborative framework for predicting RIL in cancer patients. Guided by Align-then-Aggregate strategy, AANet spatially and semantically aligns heterogeneous modalities before aggregating them to fully model intermodal interactions, effectively leveraging patient information distributed across multiple modalities to achieve RIL prediction. We discuss this framework in three parts: the necessity of multi-modal data fusion guided by Align-then-Aggregate in RIL, the effectiveness of the Align stage in eliminating modal heterogeneity, and the efficacy of the Aggregate stage in modeling intermodal interactions. Finally, we explore limitations of the current approach.

5.1. Necessity of align-then-aggregate for multi-modal fusion

The multi-modal fusion strategy of Align-then-Aggregate enables more effective utilization of patient information from multiple sources,

thereby supporting improved RIL prediction. RIL prediction depends on anatomical information from CT, dosimetric information from dose maps, and patient-specific information from clinical data. Among the single-modality settings in Table 5, CT achieves slightly better performance than dose or metadata alone, suggesting that anatomical information provides a stronger direct basis for RIL discrimination. However, the limited performance of all single-modality models indicates that neither structural anatomy, dose distribution, nor clinical variables alone are sufficient to characterize the complex risk of lymphopenia, which supports the necessity of multi-modal modeling.

The pairwise fusion results further reveal that the benefit of multi-modal learning is not uniform across modality combinations. Compared with CT-dose fusion alone, combinations involving metadata achieve stronger performance, suggesting that clinical variables provide important contextual cues for interpreting anatomical and dosimetric patterns. Moreover, while MLP improves upon simple concatenation, both remain clearly inferior to AANet in the three-modal setting, indicating that shallow fusion is insufficient to resolve inter-modal heterogeneity

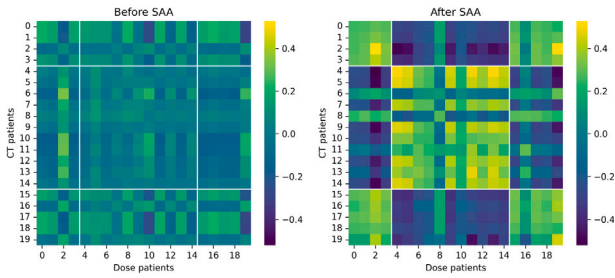


Fig. 8. Visualization of semantic alignment. Patients were stratified and sorted by category (white lines) while preserving the original class proportion, and the pairwise cross-modal similarity between CT and dose features was visualized before and after SAA.

and organize higher-order complementary interactions across CT, dose, and clinical information.

Moreover, the comparison results in Table 11 show that AANet performs better overall than existing fusion methods. This suggests that the advantage of multi-modal learning in RIL prediction does not only come from using more modalities, but also from properly handling the heterogeneity among them and effectively modeling their complementary interactions, which enables AANet to leverage information from these modalities more effectively. These findings support the necessity of an Align-then-Aggregate strategy for this task.

5.2. Effectiveness of align stage for modality heterogeneity

The Align stage is effective for mitigating the clear heterogeneity between CT and dose modalities in both semantic content and spatial representation. CT mainly provides anatomical information, whereas dose maps describe radiation distribution. Directly fusing these two modalities may limit the model's ability to capture their complementary relationships, where the Align stage is required to bridge the modality gap.

This is supported by the ablation results. Table 6 shows that introducing SAMF or SMMF improves the model performance, while combining both modules achieves the best overall results. Table 7 demonstrates that incorporating MHSA during the semantic alignment stage effectively improves performance. Table 8 demonstrates the importance of multi-directional semantic alignment in SAA, as evidenced by the performance degradation observed when cross-modal attention is absent. Furthermore, Tables 9 and 10 show that multi-level alignment and the joint use of local and global contextual filtering are more effective than partial alignment settings.

Fig. 8 further visualizes the effect of semantic alignment in SAA. It should be noted that this visualization is not intended to prove raw data-level CT-dose correspondence, because such correspondence is determined by paired patient data and spatial preprocessing. Instead, Fig. 8 visualizes the learned feature-level semantic consistency between CT and dose modalities in relation to RIL-grade discrimination. Before SAA, the pairwise cross-modal similarity is relatively scattered, indicating that CT and dose representations remain heterogeneous in the discriminative feature space. After SAA, the similarity map becomes more structured and shows clearer category-related block patterns, suggesting that CT and dose features become more consistent with respect to RIL-grade-related discriminative patterns. This indicates that SAA helps reduce semantic heterogeneity and improves the collaborative contribution of CT and dose modalities to RIL classification.

5.3. Effectiveness of aggregate stage for inter-modal interactions modeling

The Aggregate stage is effective for modeling the collaborative interactions between modalities. Although the Align stage reduces the

heterogeneity between CT and dose modalities, these aligned image features cannot be directly fused with clinical data in a balanced manner, leading to imbalanced and inadequate interaction between modalities. Therefore, an additional aggregation process is required to model the collaborative interactions between radiation-related image features and clinical factors.

This is supported by the experimental results. Table 5 shows that the best performance is achieved when CT, dose, and clinical data are jointly used for modeling by proposed Aggregate stage, indicating that clinical information provides complementary cues beyond image modalities alone. In addition, Table 6 shows that the complete model achieves the best overall performance, suggesting that clinically guided aggregation further enhances the discriminative ability of aligned multi-modal features. These results indicate that the Aggregate stage is important for inter-modal collaboration in RIL prediction.

5.4. Limitations

Although our proposed AANet has achieved better performance in the experiments and ultimately enriches the research content in the RIL prediction of radiation therapy for NPC, there are some limitations:

Dataset with limited sample size and class imbalance. This study used data from only 211 NPC patients from 2 hospitals to develop and evaluate the model, which may limit the generalizability of the proposed framework. In addition, the imbalanced class distribution in the dataset may have affected the predictive performance, especially for minority classes, as reflected by the varying class-wise performance in Fig. 6 and the relatively lower F1-score in Table 4. Moreover, G1 RIL cases were too few to be included in model training.

To further examine this issue, we additionally conducted exploratory experiments with two commonly used data-imbalance mitigation strategies, namely focal loss [33] and re-sampling [34], and the results are reported in Table 12. Both strategies led to overall performance degradation under the current setting, suggesting that directly applying simple generic remedies may be suboptimal for this task. A possible reason is that the current dataset remains limited in size and contains only a small number of minority-class samples, such that re-sampling may disturb the original data distribution, while focal loss may make optimization less stable in this small-scale multi-modal setting. In future work, more suitable imbalance-aware strategies should be further explored.

Model with limited interpretability. Due to the prevalent “black box” nature of deep learning methods, our approach still lacks transparency and explainability. In addition, our study was only used to predict the grading of RIL and did not provide other prognostic indicators. Therefore, the current model can be used to evaluate radiotherapy-related risk, but it cannot yet directly support treatment planning.

6. Conclusion

In this paper, a multi-modal collaborative network framework AANet was proposed to achieve radiation-induced lymphopenia (RIL) prediction in nasopharyngeal carcinoma (NPC) radiation therapy (RT). Based on the Align-then-Aggregate strategy, AANet achieved multi-modal interactive representation learning in a stage-wise manner: it first semantically and spatially reduced the heterogeneity between modalities through multi-level intra- and inter-modal alignment, and then modeled interactions between aligned modalities through fair joint aggregation of the 3-dimensional modalities with clinical information. Extensive experimental results have shown impressive performance of AANet in RIL prediction, demonstrating the effectiveness of its architectural design and its promising potential in NPC treatment.

In future work, we will primarily focus on two aspects: improving data availability and enhancing model interpretability. On the one hand, we will employ multi-center studies, retrospective studies, data

Table 12
Exploratory comparison of data-imbalance mitigation strategies shows that simple mitigation methods may degrade performance.

Method	Stat	AUC(↑)	ACC(↑)	PRE(↑)	SEN(↑)	SPE(↑)	F1(↑)
AANet	mean	0.8562	0.8148	0.8375	0.7670	0.8870	0.7784
	std	0.0450	0.0380	0.0183	0.0808	0.0304	0.0724
+Resample	mean	0.8472	0.7582	0.7597	0.7219	0.8650	0.7177
	std	0.0608	0.0427	0.0721	0.0832	0.0301	0.0662
+FocalLoss	mean	0.8196	0.7443	0.7368	0.6901	0.8557	0.6849
	std	0.0442	0.0526	0.0677	0.0723	0.0325	0.0627

sharing, and external validation to obtain more data, and further investigate more effective strategies for mitigating class imbalance, thus improving the reliability and generalizability of the predictive model and better exploiting multi-modal information for RIL prediction. On the other hand, we will enhance interpretability by analyzing modality-specific contributions, visualizing informative regions in CT images and dose maps, and quantifying the effects of clinical variables on model decisions. In addition, we will explore extending the current framework to multi-task prediction, with the aim of providing clinicians with more practical support for radiotherapy planning and cancer treatment.

CRediT authorship contribution statement

Rongchang Zhao: Writing – original draft, Methodology, Formal analysis, Conceptualization. **Hengzhang Deng:** Writing – original draft, Software, Methodology, Formal analysis, Conceptualization. **Zhangyue Wu:** Writing – review & editing, Visualization, Validation, Investigation, Data curation. **Zijian Zhang:** Writing – review & editing, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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